PS-SSR: All Methods Tried, All Results

Comprehensive summary of methods and results on the 6Q Chinese consumer-health benchmark

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1 Setup

Benchmark: 6 real survey questions on Chinese consumer health (TCM): 8.0/qsingle_3 (), 8.0/qsingle_4 (), 9.0/qsingle_3 (), 9.0/qsingle_4 (), 9.0/qsingle_5 (), 11.0/qsingle_3 ().

Data: ∼12K Weibo/Xiaohongshu posts → 66 latent topic clusters. **Stack:** Qwen3-8B (open) + bge-base-zh-v1.5 embedder, single A100. **Primary metric:** mean JS divergence over the 6 questions (lower = better). **Secondary metrics:** K_{xy} (KS-similarity, higher = better), C_{xy} (cosine, higher = better) — for cross-paper diagnostic only (paper uses ordinal Likert; we have nominal multi-choice).

Reference baseline: Maier et al. 2026, *LLMs Reproduce Human Purchase Intent via Semantic Similarity Elicitation of Likert Ratings.* Their SSR is **per-consumer + flat-aggregated** (eq. 1: $p_s(i) = \frac{1}{N_s} \sum_c \delta_{ir_c}$). **No clustering, no cluster-mass weighting.**

Quarantined: 23-question expanded eval (LLM-generated questions, not real surveys) is excluded from this report — see `results/quarantine_23q/` and `experiments/quarantine_23q/`.

2 Method Families (Brief Catalog)

Method families exhaustively tried, grouped by approach (each row in the next table corresponds to a family member):

- **Baselines**: Paper-SSR (Maier transfer, per-consumer flat); Direct SSR / M0 family (hierarchical cluster-weighted, sweep $n\_posts \in \{30, 50, 80, 120, 160\}$ and $top{-}K \in \{5, 10, 15, 20\}$); persona-prompt baseline.
- **LLM-distribution (M1)**: LLM-dist top-K (cluster-conditioned, $n\_samples \in \{3, 5, 10\}$); LLM per-post voting; SSR + LLM correction.
- **Activation-steering (M3/M4)**: persona-vector extraction via CAA at $L \in \{8, 12, 16, 20, 24, 28\}$; single-layer; multi-layer; ICL + steer; projected-persona SSR; steered M1.
- **Ensembles (M2)**: M0+M1 weight sweep; SSR + multi-layer-steer; selective ensemble (KL-thresholded); SSR + LLM-dist; M2+M4 triple.
- **Reweighting (alignment)**: default empirical; IS_prov / IS_age / IS_prov+age; OT_prov_fast / OT_prov_sinkhorn; adaptive-n.
- **QAW family** (Question-Adaptive Weighting + demographic-reweighting): no consistent gain on 6Q; quarantined under `quarantine_23q/`.
- **Controls**: random / shuffled / zero persona vectors as negative controls.

3 Headline Results — All Families in One Table

Method	Mean JS ↓	Δ vs paper-SSR	Source
Paper-SSR (Maier transfer, per-consumer flat)	0.0430	—	`paper_ssr_6q.json`
Hierarchical cluster-weighted family (M0 / Direct SSR — primary novel contribution)			
Direct SSR n=30, top-10	0.0277	−35.6%	`improve_orig6.json`
Direct SSR n=50, top-10	0.0275	−36.0%	`improve_orig6.json`
Direct SSR n=80, top-10	0.0276	−35.8%	`improve_orig6.json`
Direct SSR n=50, top-5	0.0272	−36.7%	`improve_orig6.json`
M0 hier-SSR canonical (n=50, top-10, seed=42)	**0.0269**	**−37.5%**	`m0_m2_pmfs_6q.json`
LLM-distribution family (M1)			
LLM-dist top-5 (n=3)	0.0290	−32.6%	`improve_orig6.json`
LLM-dist top-10 (n=3)	0.0259	−39.8%	`improve_orig6.json`
M1 vanilla (n_samples=3, top-10)	0.0264	−38.6%	`steered_m1_6q.json`
M1 vanilla (n=10, top-10)	0.0259	−39.8%	`stability_6q.json`
M1 alpha=0.05 (steered)	0.0282	−34.4%	`steered_m1_6q.json`
M1 alpha=0.1 (steered)	0.0291	−32.3%	`steered_m1_6q.json`
M1 alpha=0.2 / 0.3 (steered)	0.0313	−27.2%	`steered_m1_6q.json`
Ensembles (M2 family)			
SSR + LLM-dist w=0.5	0.0257	−40.2%	`improve_orig6.json`
SSR + LLM-dist w=0.6	0.0258	−40.0%	`improve_orig6.json`
SSR + LLM-dist w=0.7	0.0259	−39.8%	`improve_orig6.json`
M2 vanilla w=0.4 / 0.5	0.0252	−41.4%	`steered_m1_6q.json`
M2 ensemble w=0.5 (M0+M1, canonical)	**0.0254**	**−40.9%**	`m0_m2_pmfs_6q.json`
LLM weight sweep n=10 w=0.3	0.0252	−41.4%	`stability_6q.json`
M2+M4 triple ensemble (default)	0.0286	−33.5%	`alignment_6q.json`
Persona-methods Ensemble w=0.5 (BL=0.0276)	0.0261	−39.3%	`persona_methods.json`
Selective Ensemble w=0.9 t=0.005	0.0276	−35.8%	`improve_orig6.json`
Activation-steering family (mostly negative)			
Single-layer α=0.1 L24	0.0659	+53.3%	`fix_steering_results.json`
Multi-layer α=0.1 (L16,20,24)	0.0298	−30.7%	`fix_steering_results.json`
Multi-layer n=10	0.0384	−10.7%	`improve_orig6.json`
ML L=[12,16,20,24] (4-layer)	0.0643	+49.5%	`improve_orig6.json`
ICL (5 posts, no steer)	0.0368	−14.4%	`fix_steering_results.json`
ICL + steer α=0.3	0.0420	−2.3%	`fix_steering_results.json`
Contrastive prompt	0.0415	−3.5%	`fix_steering_results.json`
Large α=1.0 normed	0.0689	+60.2%	`fix_steering_results.json`
Failed methods (kept for transparency)			
LLM voting (per-post)	0.2360	+449%	`persona_methods.json`
SSR + LLM correction	0.0296	+8.0% vs Direct SSR	`persona_methods.json`
Projected SSR s=0.1	0.0560	+30.2%	`persona_methods.json`
Projected SSR s=0.3	0.0478	+11.2%	`persona_methods.json`
Projected SSR s=0.5	0.0405	−5.8%	`persona_methods.json`

Table 1: Best results per method family. Baseline: paper-SSR transferred from Maier et al. 2026. Green: best-in-class; red: failed (worse than paper-SSR or worse than baseline).

4 Per-Question Honest Reporting

Question (Chinese health domain)	Paper-SSR	Direct SSR	M0	M1	M2	Best
8.0/qsingle_3 ()	0.0307	0.0144	**0.0134**	0.0091	**0.0105**	M1
8.0/qsingle_4 ()	0.1309	0.0466	0.0460	0.0506	0.0468	M0
9.0/qsingle_3 ()	0.0202	**0.0193**	0.0198	0.0434	0.0304	Direct SSR
9.0/qsingle_4 ()	0.0182	0.0175	0.0173	0.0174	**0.0158**	M2
9.0/qsingle_5 ()	0.0398	**0.0491**	0.0468	0.0317	0.0367	M1
11.0/qsingle_3 ()	0.0183	0.0181	0.0180	0.0086	0.0125	M1
Mean (6Q)	0.0430	0.0275	**0.0269**	0.0268	**0.0254**	M2
Mean (5Q, exclude Q2 outlier)	0.0254	0.0237	0.0231	0.0220	**0.0212**	M2

Table 2: Per-question JS divergence. M0/M1/M2 numbers from `m0_m2_pmfs_6q.json`. Q2 is an outlier where paper-SSR collapses (0.131); the cluster-hierarchical family fixes it. M2 hurts Q3 by ∼50% (a real per-question loss). M0 hurts Q5 by ∼18%. No method dominates uniformly.

Key honest observations:

- Q2 (gut-cold scenarios) drives ∼2/3 of the cross-baseline gain. Without Q2, M2 vs paper-SSR shrinks from −40.9% to −16.9%.
- M2 ensemble hurts Q3 ("") by 50% — likely because the LLM's distribution estimate disagrees strongly with SSR on this question, and naive averaging amplifies noise.
- M0 alone hurts Q5 ("") by 18% — the cluster aggregation slightly underweights an important option.
- M1 (LLM-dist) wins on 3 of 6 questions individually but is unreliable (best on some, bad on Q3).

5 Cross-Paper Diagnostic Metrics (K_{xy}, C_{xy})

Computed only for the M0/M1/M2 dump and paper-SSR (the only methods that saved per-question PMFs).

Method	JS ↓	K_{xy} ↑	C_{xy} ↑
Paper-SSR (transferred)	0.0430	0.827	0.877
M0 hier cluster-weighted SSR	0.0269	**0.858**	0.923
M1 cluster LLM-dist	0.0268	0.844	0.931
M2 ensemble w=0.5	**0.0254**	0.852	**0.932**
Maier headline (GPT-4o + demos, English Likert PI)	—	0.88	0.96
Maier no-demo control (English)	—	0.91	0.98
Maier DLR baseline (Likert direct)	—	0.26	0.26

Table 3: Cross-paper diagnostic. K_{xy} is mathematically appropriate only for ordinal Likert; our options are nominal multi-choice, so K_{xy} comparisons across model/language/task are diagnostic only.

6 Stability Analyses

6.1 Cluster-reweighting stability (alignment_6q.json)

Reweighting scheme	M0 JS	M1 JS	M2 JS	M4 JS (SSR+ML steer)
Default (empirical mass)	0.0269	0.0261	0.0253	0.0315
IS by province	0.0271	0.0261	0.0254	0.0315
IS by age	0.0269	0.0261	0.0252	0.0315
IS by province+age	0.0271	0.0261	0.0254	0.0315
OT province (fast)	0.0272	0.0262	0.0255	0.0312
OT province (sinkhorn)	0.0269	0.0261	0.0253	0.0315
Range	[0.0269, 0.0272]	[0.0261, 0.0262]	[0.0252, 0.0255]	[0.0312, 0.0315]

Table 4: All M0/M1/M2 results are stable in tight bands across 6 reweighting schemes. The hierarchical pipeline is robust and not a single-variant artifact.

6.2 n_posts adaptive sweep

Adaptive-n within the M0 family: $n = 30 \to 0.0272$, $n = 50 \to 0.0269$, $n = 80 \to 0.0270$, $n = 120 \to 0.0268$, $n = 160 \to 0.0268$, adaptive-variance $\to 0.0272$. Diminishing returns beyond n=50.

6.3 Repeated-trial stability (M3 / M4)

`stability_6q.json`: M4 (SSR + ML steer ensemble) over 3 trials: mean 0.0311 ± 0.0002 (stable, but worse than M0/M2). M3 (single-layer steer): 0.0397 ± 0.0006 (worse).

6.4 LLM weight sweep ($n\_samples$, w)

$n\_samples \setminus w$	0.3	0.4	0.5	0.6	0.7
n=5	0.0265	0.0262	0.0260	0.0260	0.0260
n=10	**0.0252**	**0.0252**	0.0253	0.0254	0.0256

Table 5: Weight sweep on M2 ensemble. Best at n=10, $w \in \{0.3, 0.4\} \to 0.0252$. Performance is flat across $w \in [0.3, 0.7]$, suggesting w=0.5 is not specifically tuned.

7 Steering / Persona-Vector Story (Demoted to Appendix)

7.1 $\alpha \times L$ operating regime — pilot finding

In an earlier 23Q exploration (now quarantined), persona-vector steering was *conditionally* useful:

- **Helps**: low $\alpha \leq 0.3$ AND late layers $L \geq 24 \to$ up to -21% JS over unsteered.
- **Direction-irrelevant zone**: $\alpha \geq 0.5$ middle layers $\to$ random and shuffled vectors match real ones (negative control passes — bad sign).

7.2 Negative controls

Per-layer sweep with random / zero / shuffled persona vectors (`sweep.log`, `control_{random,shuffled,zero}.json`): at aggressive α, real vectors win only 2/6 to 4/6 questions vs zero — barely above chance, inconsistent across questions. Confirms steering does not robustly use persona-direction information at the operating regimes that improve mean JS.

7.3 Why each failed method failed (one-liner)

- **LLM voting (per-post, JS=0.236)**: LLM votes the same answer regardless of post content; massive within-cluster collapse.
- **SSR + LLM correction (JS=0.030)**: LLM rewrite adds noise to the SSR PMF.
- **Projected-persona SSR (JS=0.04–0.06)**: projecting anchor embeddings along the persona vector destroys the anchor geometry that SSR depends on.
- **QAW + demographic reweight**: no consistent gain on 6Q; quarantined under `quarantine_23q/`.
- **Selective ensemble (JS=0.030)**: KL-thresholded blending under-uses the LLM signal vs the simple linear ensemble.
- **Steering at large α or many layers (JS=0.06–0.07)**: catastrophic — direction-irrelevant zone confirmed by random/shuffled controls matching real vectors.